

Administrative markers of maternal near miss in Brazil: regional patterns, age composition and socioeconomic associations in SUS hospital admissions, 2002–2025

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Abstract

Objective: To describe administrative markers of maternal near miss in obstetric admissions financed by Brazil's Unified Health System (SUS), examining regional and age patterns, agreement between diagnostic lists and socioeconomic associations.

Methods: We analyzed SIH-SUS admissions processed in 2002–2025 among females aged 10–49 years with obstetric specialty or a Chapter O diagnosis and known discharge outcome, without requiring prenatal registration. Surviving admissions were identified using Waterstone-derived, Mantel-derived and WHO-related principal-diagnosis lists. We calculated crude and age-standardized frequencies per 1,000 obstetric admissions and exploratory bivariate socioeconomic correlations. A complementary analysis applied the diagnostic approach of Rosendo and Roncalli to residents aged 15–49 in 1,794 Northeastern municipalities, using admissions in 2008–2012 and Census 2010 indicators.

Results: Across 58,376,917 admissions in 287 nationally complete months, Waterstone-CID, Mantel-CID and WHO-CID frequencies were 36.90, 9.29 and 21.09 per 1,000 admissions. The North had the highest pooled frequencies, including after age standardization. Frequencies were higher at ages 35–49 than at 10–34. Prenatal registration was present in 16.47% of admissions; restriction to that subgroup reduced frequencies and changed regional rankings. Of 90 regional Pearson comparisons, 19 had multiplicity-adjusted $q < 0.05$; seven changed direction after first differencing. The Northeastern complement identified 140,519 of 3,913,500 admissions (35.91 per 1,000), with variation in diagnostic composition and municipal socioeconomic associations

across states.

Conclusion: Public SUS records reveal substantial regional and age variation in administrative markers. Population selection and diagnostic definition materially affect their interpretation. These findings support transparent administrative surveillance and hypothesis generation, while requiring distinct clinical and population-based evidence to explain underlying risks.

Keywords: maternal near miss; severe maternal morbidity; hospital information systems; health inequalities; Brazil; public health surveillance.

Introduction

Maternal near miss describes survival following a life-threatening complication during pregnancy, childbirth or the postpartum period. Studying survivors complements maternal mortality surveillance by identifying a larger group of severe events and opportunities to investigate care pathways [1]. The clinical frameworks proposed by Mantel, Waterstone and the World Health Organization (WHO) differ in their emphasis on conditions, organ dysfunction and management criteria [1, 2, 3]. Their application requires information that is not uniformly available in routine administrative records.

In Brazil, the Hospital Information System of SUS (SIH-SUS) provides public records of publicly financed hospital care over several decades. These data offer substantial geographic and temporal reach. Studies by Rosendo and Roncalli, Carvalho and colleagues, and Herdt and colleagues have used administrative definitions to investigate maternal morbidity and its territorial distribution [4, 5, 6]. Their contribution is to describe patterns detectable in the available information system; diagnosis codes alone do not establish the clinical thresholds required for every near-miss criterion. Hospital comparisons with medical records and comparisons between clinical definitions demonstrate why administrative identification and clinical confirmation must remain distinct [7, 8].

Two questions are particularly relevant to interpreting administrative patterns. First, regional and temporal differences may reflect changes in the age distribution of hospitalized women as well as changes in the recorded conditions. Crude frequencies do not separate those components. Second, imposing a requirement for a completed administrative field may select a subgroup whose relationship to the broader obstetric population varies across place and time. Prenatal-registration fields warrant this consideration because the presence of a registration number is not

equivalent to attendance or quality of prenatal care.

The geographic scale of analysis also matters. Municipal comparisons of socioeconomic context address a different question from correlations between annual regional averages. Rosendo and Roncalli's ecological study in Rio Grande do Norte (RN) provides a methodological reference for examining local heterogeneity [4], while population surveys contribute evidence on individual and household circumstances [9]. These forms of evidence are complementary and should not be treated as interchangeable estimates of the same association.

This study describes administrative markers of maternal near miss in obstetric admissions financed by SUS between 2002 and 2025. We compare three diagnostic lists, examine regional frequencies before and after age standardization, characterize diagnostic composition and list overlap, and assess sensitivity to requiring prenatal registration. We also explore regional socioeconomic associations and perform a historical municipal analysis across the nine Northeastern states inspired by Rosendo and Roncalli.

Methods

Design, sources and population

We used publicly available reduced SIH-SUS hospital files for processing months from January 2002 through December 2025, obtained from DATASUS on 16 September 2026 [10]. Files from subsequent processing years were excluded. The analytical unit was a hospital admission authorization (AIH), rather than a uniquely identified woman. Time was defined by processing competence, which may differ from the date of the clinical event.

Eligibility required a normal AIH, a valid identifier, female sex, age 10–49 expressed in years, and a recorded discharge outcome. An admission was considered obstetric when it had obstetric specialty or a Chapter O diagnosis in any available diagnostic field. A non-zero prenatal-registration number was not required. Admissions ending in death were included in the denominator but could not be near-miss identifications. The analysis therefore describes surviving admissions with markers among all eligible obstetric admissions.

When an AIH appeared more than once, we retained its latest observed processing competence within the study period, including later versions that no longer satisfied eligibility. Identical analytical records at that competence counted once; conflicting records were excluded and quantified. Without a person-level linkage key, separate admissions of the same woman could

not be identified.

Administrative definitions and complementary attributes

Principal diagnoses were matched against three ICD-10 lists derived from the Brazilian administrative applications of the Waterstone, Mantel and WHO frameworks [4, 6, 7, 8]. They contained 102, 158 and 262 distinct codes, respectively. Throughout this article, Waterstone-CID, Mantel-CID and WHO-CID denote these administrative lists, not clinical confirmation of the corresponding frameworks. Matching was exact after removal of punctuation; three-character categories were not automatically expanded to all subcategories. The complete codebook is supplied with the analytical materials.

An admission could belong to more than one list. We reported their union and the observed mutually exclusive positive combinations. Pairwise similarity was assessed with the Jaccard index, defined as the size of the intersection divided by the union; this measures overlap, not diagnostic validity. Diagnostic groups describe recorded conditions and do not establish severity thresholds for individual admissions. Recorded intensive-care use was defined by at least one ICU day; absent information remained unknown.

We evaluated three definition sensitivities: restricting the same obstetric population to records with a non-zero prenatal-registration number; identifying markers in any available diagnostic field; and using only surviving admissions as the denominator. Prenatal-field completeness was interpreted as a recording characteristic. Secondary-diagnosis availability and coding practices may change over time.

Geography, coverage and age standardization

The national analysis used state of processing/hospitalization and the five IBGE geographic regions. The expected inventory comprised 26 states and the Federal District, comprising 27 federative units, 24 years and 12 months, totaling 7,776 files. Amapá/October 2007 was not listed in the source. Pooled regional comparisons therefore used only the 287 processing months complete in all 27 federative units. Annual series used complete months within each territory; years with fewer than 12 months were identified as partial and excluded from annual plots and temporal correlations. No missing month was treated as zero or annualized.

Frequencies were expressed as identifications per 1,000 eligible obstetric admissions. These are not WHO maternal near-miss ratios per live births. For direct age standardization, we used eight

five-year groups from 10–14 to 45–49 years. The standard weights were the age distribution of all national eligible admissions in the common-month pooled analysis. Each territory-year frequency was the weighted sum of its age-specific frequencies using these fixed weights. A descriptive frequency ratio compared ages 35–49 with 10–34, with a conventional log-scale 95% confidence interval (CI).

Regional socioeconomic associations

Six official IBGE series were obtained through the aggregate-data API [11]. PNAD series described mean nominal monthly income among persons aged 10 years or older, the percentage of that population with no schooling or up to five years of schooling, and the percentage of permanent private households with a sewer connection or septic tank. PNAD Contínua series described real mean monthly income among residents with income, the percentage of persons aged 25 years or older with no schooling or fewer than five years of schooling, and sewer/septic provision among households with a bathroom, toilet or latrine. The income series differed in population universe and monetary basis; nominal PNAD income was not interpreted as purchasing power. PNAD Contínua income used the 2025 price basis of the downloaded series.

These series were analyzed separately because their universes and definitions differed. Records were joined by region code and year, without interpolation. PNAD comparisons for the North began in 2004, after expansion of its rural coverage [12]. Analyses included only years with complete regional SIH coverage and an observed covariate.

For each indicator, region and list, we estimated Pearson correlations with Fisher 95% CIs, Spearman correlations, and Pearson correlations of first differences between consecutive years. At least four nonconstant paired observations were required. Benjamini–Hochberg adjustment was applied to the family of 90 Pearson tests using crude frequencies. The 90 age-standardized comparisons formed a separate sensitivity family. Conventional CIs and p-values assume independent observations and do not account for serial dependence; the analyses were exploratory. No multivariable models or causal effects were estimated.

Historical municipal analysis in the Northeast

The complementary ecological analysis extended the diagnostic approach of Rosendo and Roncalli [4] to all nine Northeastern states. We selected normal AIHs for female residents aged 15–49 with known discharge outcome, processed in 2008–2012 and hospitalized anywhere in Brazil. Eligibility required a Chapter O principal diagnosis or an obstetric procedure,

operationalized through SIGTAP prefixes 030310, 0310 and 0411; prenatal registration was not required. Residence codes linked admissions to municipalities. We resolved AIH versions using the full 2002–2025 archive before selecting the historical processing period.

The diagnostic-code table of Rosendo and Roncalli, drawing on Sousa and colleagues [13], defined the overall marker set and its hypertensive and hemorrhagic subsets. This set excludes the HIV and acute-abdomen groups present in the national Waterstone-CID list. Surviving admissions contributed to numerators; all eligible admissions contributed to denominators. Municipal and pooled state frequencies were crude. This complement therefore differs from the national analysis in period, age limits, obstetric eligibility, diagnostic list and residence-based geography.

Municipal frequencies were paired by IBGE code with Census 2010 income per capita, illiteracy among persons aged 10 or older, and sewer/septic coverage [11]. The historical period kept admissions close to the covariate year. Pearson and Spearman correlations gave equal weight to each municipality and were calculated for each state and the Northeast. Sensitivities required at least 1,000 admissions and/or restricted hospitalization to the state of residence. Within each of these four combinations, Benjamini–Hochberg adjustment was applied separately to 90 Pearson and 90 Spearman tests ($10 \text{ territories} \times 3 \text{ outcomes} \times 3 \text{ indicators}$). There were no outcome-based exclusions. Correlations after subtracting state-specific means from both variables described within-state associations for the pooled Northeast.

In addition to conventional Fisher CIs and p-values, we estimated percentile 95% CIs for Pearson correlations using 2,000 bootstrap resamples of whole historical IBGE microregions, stratified by state for the pooled Northeast. The 1,794 municipalities belonged to 188 microregions. These exploratory intervals allow within-block dependence but do not address dependence between blocks and were not adjusted for multiple comparisons. Municipalities replaced the spatial clusters of the original study, and the three Census indicators did not reproduce its full contextual variable set; this was not an exact replication. An additional exploratory decomposition by principal diagnosis was specified after inspecting the main state results and before tabulating individual codes, without changing the marker set or eligibility.

Reproducibility

The analysis plan, full code lists, source metadata, coverage inventory, Python scripts and aggregate results accompany the manuscript in the working repository. Automated checks

reconcile monthly totals with age strata, regional totals with national totals, and positive identifications with their denominators and list intersections. A static public-facing dashboard presents the aggregate results and definitions. No individual microdata are redistributed with the dashboard.

Results

Coverage and study population

All 7,775 available files were processed, representing 288,381,354 hospital record versions before eligibility and deduplication. AP/October 2007 was absent from the source inventory. The common-month analysis comprised 58,376,917 eligible obstetric AIHs in 287 months, including 20,566 admissions ending in death. Non-zero prenatal registration was present in 9,613,424 admissions (16.47%).

Diagnostic lists and regional frequencies

Waterstone-CID identified 2,154,047 admissions (36.90 per 1,000); Mantel-CID identified 542,154 admissions (9.29 per 1,000); WHO-CID identified 1,231,310 admissions (21.09 per 1,000). Their union included 2,271,149 admissions. Table 1 presents crude and age-standardized regional frequencies. The North had the highest pooled frequency for each list and the South the lowest; this ordering persisted after age standardization. The pooled ordering did not describe every year: by 2025, the Northeast had the highest Waterstone-CID frequency (Figure 1). Nationally, Waterstone-CID increased from 28.74 per 1,000 in 2002 to 45.65 in 2025; the corresponding standardized frequencies were 29.66 and 44.23. Mantel-CID frequencies were 11.46 and 11.01, and WHO-CID frequencies were 21.06 and 21.21, at those endpoints. These endpoint comparisons are descriptive and do not constitute tests of temporal trend.

Table 1. Crude / age-standardized frequencies per 1,000 obstetric admissions, 2002–2025.

The same 287 nationally complete months contribute to every region. W = Waterstone-CID; M = Mantel-CID. Fixed weights are the national pooled age distribution.

Region	Admissions	W	M	WHO-CID
Brazil	58,376,917	36.90 / 36.90	9.29 / 9.29	21.09 / 21.09
North	6,805,084	42.06 / 43.32	11.89 / 11.98	29.24 / 29.85

Region	Admissions	W	M	WHO-CID
Northeast	18,872,990	40.58 / 40.85	9.25 / 9.27	20.89 / 20.96
Southeast	20,851,674	34.20 / 33.76	8.67 / 8.64	19.47 / 19.32
South	7,356,343	31.90 / 31.37	8.13 / 8.08	18.70 / 18.49
Central-West	4,490,826	34.32 / 34.73	10.28 / 10.36	21.07 / 21.27

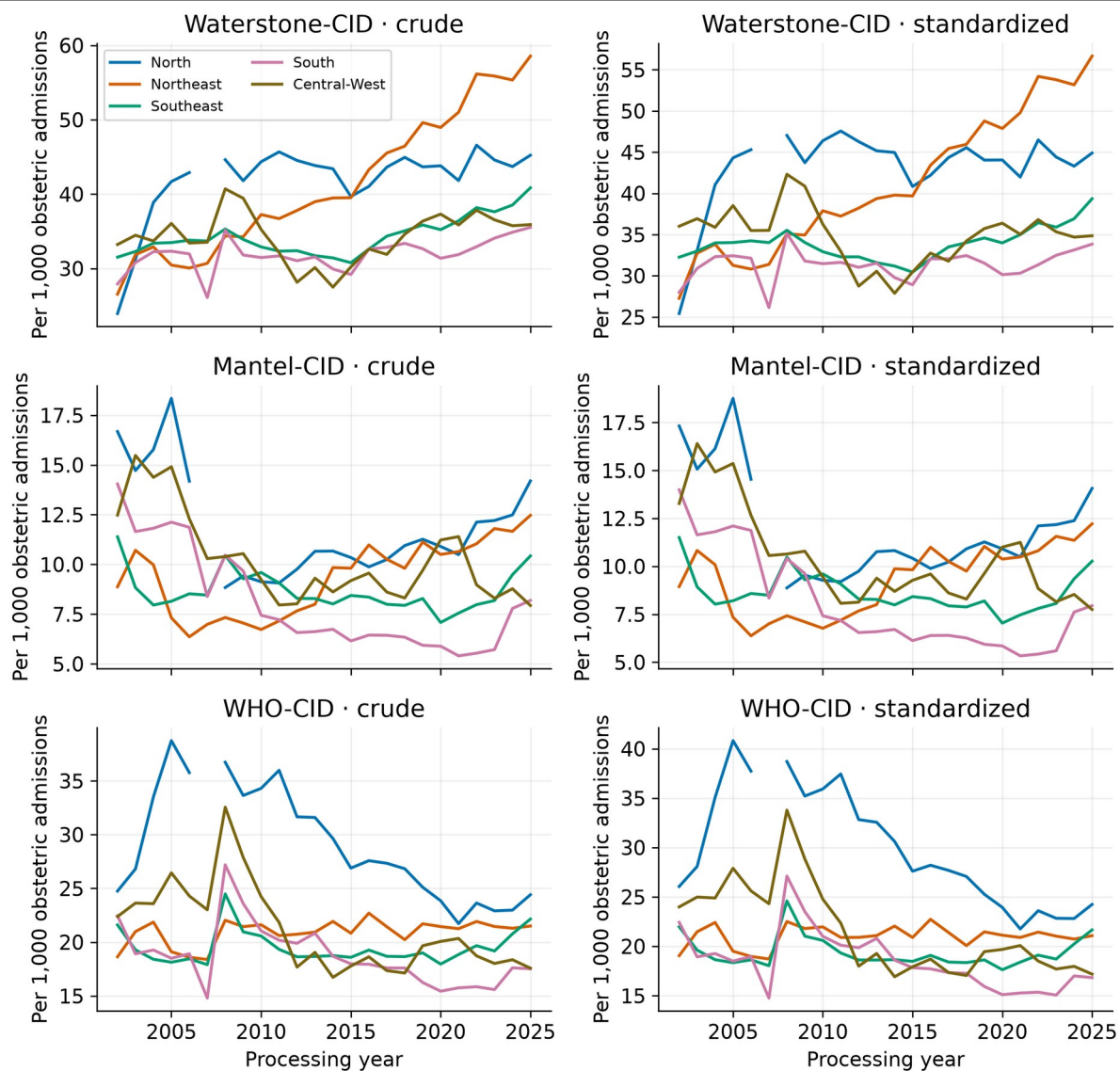


Figure 1. Regional frequencies before and after direct age standardization. Annual curves require 12 complete processing months; the North has a gap in 2007. All panels use linear scales, with panel-specific ranges. Standardization uses eight five-year age groups and the same national weights in every year.

Age, diagnostic composition and list overlap

All three national age profiles were higher at ages 10–14 than at 15–19, and rose across the older adult groups. At ages 45–49, frequencies reached 73.07 per 1,000 for Waterstone-CID, 24.85 for Mantel-CID and 51.98 for WHO-CID. The share of all eligible admissions at ages 35–49 increased from 8.40% in 2002 to 15.08% in 2025. Table 2 gives the descriptive comparison of admissions at ages 35–49 and 10–34 years. Figure 2 displays the full age profiles. Pre-eclampsia/superimposed-hypertension codes accounted for 1,038,088 (48.19%) of Waterstone-CID identifications; adding eclampsia brought the hypertensive group to 1,198,073 (55.62%). Hemorrhage-related codes accounted for 498,679 (23.15%) and infection-related codes for 437,073 (20.29%).

Table 2. Identifications by age and frequency ratios, 2002–2025. Numerators count surviving admissions with the marker; denominators include all eligible obstetric admissions in each age category. CIs are conventional and do not account for repeat admissions of the same woman.

List	10–34 years, n/N	35–49 years, n/N	Frequency ratio (95% CI)
Waterstone-CID	1,793,839/51,891,828	360,208/6,485,089	1.61 (1.60–1.61)
Mantel-CID	464,435/51,891,828	77,719/6,485,089	1.34 (1.33–1.35)
WHO-CID	1,041,312/51,891,828	189,998/6,485,089	1.46 (1.45–1.47)

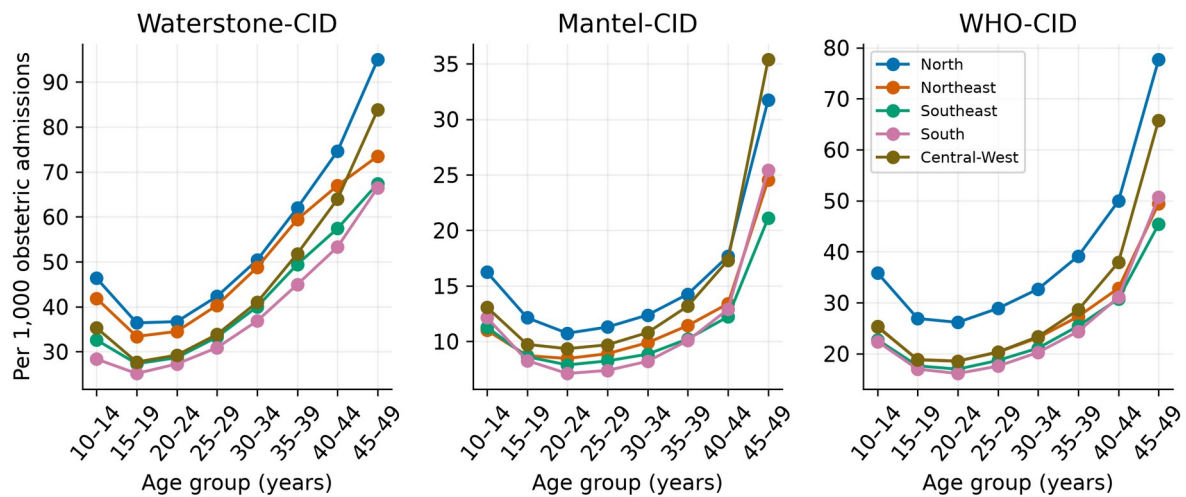


Figure 2. Age-specific frequencies by region and diagnostic list. Pooled estimates use the 287 common processing months. The horizontal axis represents five-year age groups; frequencies are specific to each group and are not standardized.

Recorded ICU use among identified admissions with known ICU status was 4.08% for Waterstone-CID; 1.74% for Mantel-CID; 2.85% for WHO-CID. The full denominators and unknown counts are supplied in the aggregate tables. The three-way intersection comprised 455,700 admissions. Waterstone-CID alone accounted for 1,039,809, whereas almost all Mantel-CID identifications also met WHO-CID: only 30 were outside that list. Jaccard similarity was 0.203 for Waterstone-CID/Mantel-CID, 0.491 for Waterstone-CID/WHO-CID and 0.440 for Mantel-CID/WHO-CID (Figure 3).

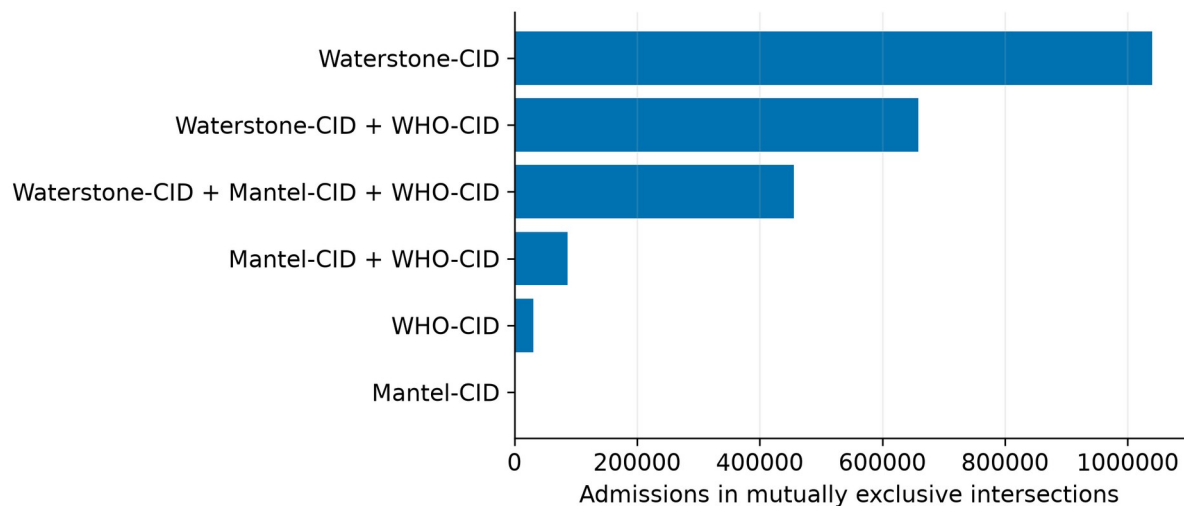


Figure 3. Mutually exclusive positive combinations of the three diagnostic lists. Each admission appears in one bar. The bars together represent the union of positive identifications, not the sum of list-specific totals.

Sensitivity to population and diagnosis fields

Among admissions with non-zero prenatal registration, frequencies were 17.44 per 1,000 for Waterstone-CID, 0.99 per 1,000 for Mantel-CID, 6.12 per 1,000 for WHO-CID. Prenatal-field completeness varied from 9.82% in the Northeast to 29.58% in the South. Within the registered subgroup, the Northeast had the highest Waterstone-CID frequency, while the South had the highest Mantel-CID and WHO-CID frequencies. This contrasted with the North's highest pooled frequencies in the full obstetric population (Figure 4). Using any available diagnosis instead of the principal diagnosis yielded 39.31 per 1,000 for Waterstone-CID, 9.88 per 1,000 for Mantel-CID, 22.42 per 1,000 for WHO-CID. Restricting denominators to surviving admissions yielded 36.91 per 1,000, 9.29 per 1,000, 21.10 per 1,000, respectively.

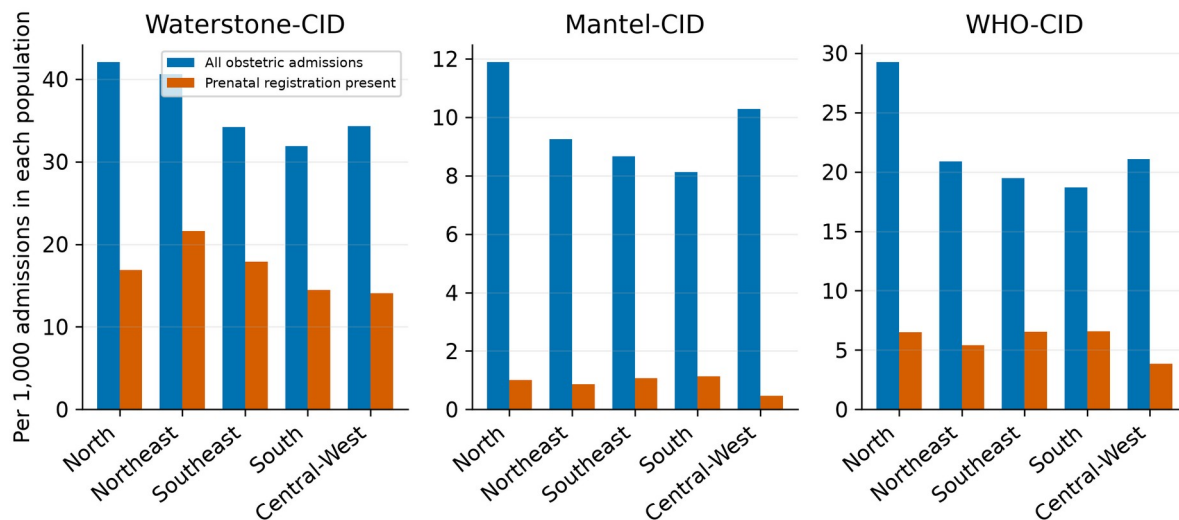


Figure 4. Frequencies in the full obstetric population and the subgroup with a recorded prenatal-registration number. Numerators and denominators use the same subgroup, region and 287 common months. Both sets are crude frequencies. Registration is not a measure of prenatal-care attendance.

Regional socioeconomic associations

Of 90 crude-frequency Pearson comparisons, 19 had BH-adjusted $q < 0.05$. The separate age-standardized sensitivity family had 25 such comparisons. Pairwise sample sizes ranged from 5 to 14 years. Among the 19 crude-frequency findings with $q < 0.05$, seven reversed sign after first differencing. For example, the Northeast correlation between nominal PNAD income and Waterstone-CID was $r = 0.932$ in levels and $r = 0.047$ in consecutive differences. Some associations remained appreciable: PNAD Contínua real income and Mantel-CID had correlations of 0.911 in levels and 0.791 in differences in the South, compared with -0.684 and -0.619 in the Central-West. Thus, neither direction nor sensitivity to temporal transformation was uniform across regions. All coefficients, confidence intervals, multiplicity-adjusted values, years and paired observations are available in the supplementary tables.

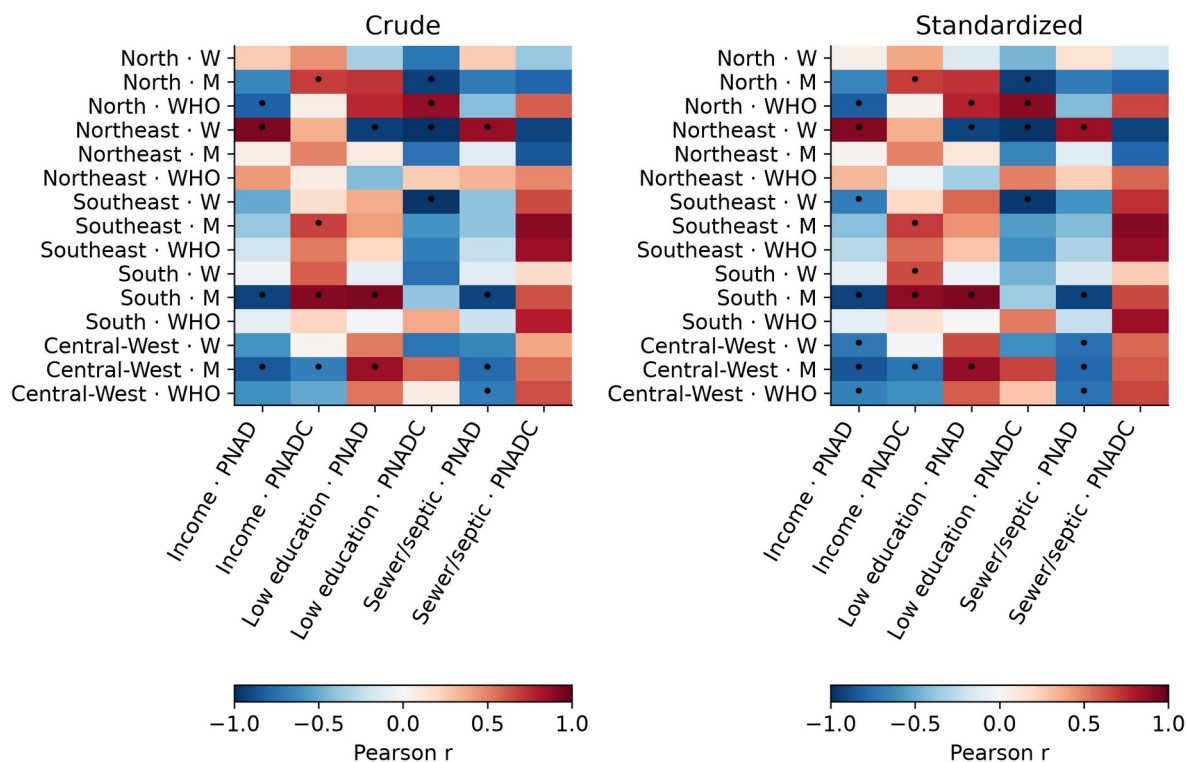


Figure 5. Pearson correlations between regional socioeconomic indicators and administrative frequencies. Left: crude frequencies; right: age-standardized frequencies. Each panel is a separate family of 90 comparisons. Dots indicate BH-adjusted $q < 0.05$. PNAD and PNAD Contínua (PNADC) series have different definitions and are not concatenated. Conventional inference does not account for temporal autocorrelation.

Historical municipal analysis in the Northeast

All 1,620 expected state-month files for 2008–2012 were available. The complement included 3,913,500 admissions from all 1,794 Northeastern municipalities and identified 140,519 surviving admissions with Rosendo-list markers (35.91 per 1,000). Hypertensive and hemorrhagic conditions accounted for 52.67% and 29.50% of identifications. State frequencies ranged from 23.27 per 1,000 in Maranhão to 72.79 in Sergipe (Supplementary Table S1 and Figure S2). In Sergipe, O03.1 (incomplete spontaneous abortion with excessive or delayed hemorrhage) accounted for 8,668 records, 95.43% of hemorrhagic and 76.88% of overall identifications. In Piauí, O06.1 (incomplete abortion of unspecified type with excessive or delayed hemorrhage) accounted for 6,142 records, 77.84% of hemorrhagic identifications.

Across all Northeastern municipalities, the overall marker frequency had a small positive correlation with income ($r = 0.110$; microregion-bootstrap 95% CI 0.056–0.171), which

remained positive after state demeaning ($r = 0.120$). Correlations with illiteracy ($r = -0.028$; CI -0.095 – 0.033) and sewer/septic coverage ($r = 0.063$; CI -0.005 – 0.135) had block intervals spanning zero. Associations varied descriptively across states: for income and overall frequency, RN had $r = 0.004$ and Sergipe $r = 0.426$ (CI -0.122 – 0.644). In Sergipe, illiteracy had $r = -0.787$ (CI -0.900 to -0.398); the inverse direction persisted in the sensitivity samples. In the primary family of 90 comparisons, 19 Pearson and 24 Spearman tests had $q < 0.05$, while 21 unadjusted block intervals excluded zero. These criteria are not interchangeable; for example, pooled sewer/septic coverage had Pearson $q = 0.037$ but Spearman $q = 0.057$ and a block interval spanning zero.

The threshold of 1,000 admissions retained 842 municipalities (46.93%) and 87.63% of admissions; 18 of 90 Pearson coefficients changed sign, some near zero. Hospitalizations outside the state of residence represented 1.38% of admissions; excluding them changed the pooled frequency to 35.44 per 1,000 and altered some municipal associations. In Pernambuco, the income–hemorrhage correlation changed from 0.044 to 0.253. Full coefficients, block intervals and all four sensitivity combinations are supplied in Supplementary Tables S2–S4; Figures S3–S4 display the municipal patterns.

Discussion

Across more than 58 million obstetric admissions, administrative near-miss markers showed substantial variation by diagnostic list, region and age. The North had the highest pooled frequencies under all three definitions, and the South the lowest, after standardization as well as before it. The national Waterstone-CID frequency was higher in 2025 than in 2002 even after accounting for age composition, whereas Mantel-CID and WHO-CID had similar endpoint magnitudes. Hypertensive diagnoses accounted for more than half of Waterstone-CID identifications. These patterns identify dimensions of maternal morbidity visible in public hospital data without equating diagnosis codes with clinically confirmed near miss.

Prenatal-registration selection was especially consequential. Requiring a non-zero registration number retained only about one sixth of eligible obstetric admissions and changed both marker frequencies and regional rankings. Because completeness differed between regions, this restriction described different selected populations across the country. The contrast is evidence about case ascertainment and administrative selection; it cannot establish a protective effect of prenatal care or distinguish nonattendance from an absent registration number.

Relationship to Brazilian and international evidence

Administrative studies provide the closest comparison for the present results. Carvalho and colleagues analyzed national Waterstone-based patterns in 2000–2012, while Herdt and colleagues studied regional variation in SIH admissions in 2010–2018 [5, 6]. Differences in eligibility, diagnosis fields, procedures, age limits and denominators prevent direct interpretation of differences in absolute frequencies as changes in clinical disease burden. The useful comparison concerns the pattern of recorded conditions and inequalities within clearly specified administrative populations.

The Northeastern complement preserved the diagnostic approach, age limits and study period of Rosendo and Roncalli [4], while examining municipalities directly across nine states. Rosendo and Roncalli analyzed 63 spatial clusters after excluding two outlying clusters and used population-weighted municipal frequencies within clusters. Our state frequencies pooled admissions, and municipal correlations used equal weights without outcome-based exclusions. The original mean of 36.67 per 1,000 is therefore not directly comparable with our pooled Northeast frequency of 35.91 or RN frequency of 46.60. Different geographic aggregation, contextual variables and weighting prevent interpreting the extension as a replication or refutation of that study.

The broader geographic coverage shows that the RN pattern cannot be assumed to describe the entire Northeast. Hypertensive predominance overall coexisted with concentrations of hemorrhagic codes in Sergipe and Piauí, and income correlations varied across states. These observations may reflect a mixture of clinical case mix, access and coding practices. They do not establish why particular codes predominated or whether those codes explain the socioeconomic correlations. The inverse association with illiteracy in Sergipe does not imply individual protection; municipal covariates describe ecological context, and the frequencies are not adjusted for age or care provision. The inclusion threshold and the location of hospitalization also affected the associations, showing why their interpretation must retain the specified admission population.

Other Brazilian ecological studies reinforce the need to distinguish sources and geographic scales. Curan and colleagues analyzed state notifications across 399 municipalities in Paraná in 2018–2021 and reported limited socioeconomic and care associations [14]. Correa and colleagues combined SIH, mortality and live-birth data for Rio de Janeiro in 2014–2016, using bivariate spatial correlations and distinguishing near miss from potentially life-threatening

conditions [15]. Their live-birth denominators and ascertainment methods differ from ours; their numerical frequencies are not direct benchmarks. Together, these studies frame questions about territorial heterogeneity rather than supplying a uniform socioeconomic relationship that should necessarily appear in every administrative definition.

Clinical evidence answers a different but essential question. The Birth in Brazil survey associated maternal near miss with individual and care-related factors [16], while hospital studies show that the performance of administrative identification depends on the information available and the clinical reference used [7, 8]. Our analysis does not estimate sensitivity or specificity against a clinical standard. Nor does agreement between two overlapping lists establish accuracy: common codes can produce high overlap even when both definitions lack the clinical information needed to classify severity.

International evidence likewise emphasizes definition and context. Firoz and colleagues documented substantial variation in the application and adaptation of near-miss criteria across settings [17]. The prospective Latin American study of organ-dysfunction definitions by Rojas-Suarez and colleagues used clinical observations that are not available in reduced SIH records [18]. These studies support transparent administrative surveillance alongside clinical investigation; their populations and denominators do not provide a universal numerical benchmark for the present frequencies. Recent Brazilian work on state surveillance [19], and proposals for national surveillance and clinical data collection [20, 21], identify possible routes toward linking these complementary forms of evidence.

Interpretation and limitations

The increased proportion of admissions at older ages contributed to differences between crude and standardized frequencies, but age composition did not account for the entire Waterstone-CID endpoint increase or the pooled regional ordering. The higher frequencies among both the youngest adolescents and older women merit more focused study; they concern hospitalized populations and cannot be translated directly into pregnancy-level risks. The very narrow conventional confidence intervals for age contrasts reflect the large number of records, not the absence of measurement error or selection.

Differences between diagnostic lists were structural as well as numerical. The near-complete inclusion of Mantel-CID identifications within WHO-CID shows why these outputs should not be interpreted as independent confirmations of severity. Similarly, principal-diagnosis

frequencies were lower than frequencies based on any recorded diagnosis, but the latter also depend on the availability and completeness of secondary fields. We retained the principal-diagnosis definition for the main comparisons and reported the broader definition as a sensitivity analysis.

Socioeconomic correlations must be interpreted at their measured scale. A regional correlation between annual averages cannot identify an individual woman's exposure or risk. Secular changes in income, education, sanitation, access and coding can generate correlations even without a direct relationship between the two variables studied. Separate survey definitions, conventional uncertainty estimates and few annual observations further limit inference. First differences probe sensitivity to shared trends but do not remove all time-varying confounding or autocorrelation. Municipal comparisons additionally depend on geographic aggregation. Block resampling acknowledges clustering but does not eliminate spatial dependence between microregions, and its intervals were not multiplicity-adjusted. Differences in coefficients between states were descriptive, without a formal interaction test. The historical municipal analysis does not describe contemporary socioeconomic conditions or extend to the other four Brazilian regions.

The age-standardized analysis addresses one component of changing admission composition. It does not adjust for parity, comorbidity, referral severity, private-sector use, availability of intensive care or other determinants. Hospital-based geographic frequencies can be affected by cross-border referral. Recorded ICU use reflects both illness and access to beds and does not itself establish organ dysfunction.

The denominator is hospital admissions financed by SUS, so changes can reflect the composition and volume of obstetric hospital care rather than the risk among all pregnancies. Repeat admissions of the same woman remain possible. Processing dates and later submissions may shift the apparent timing of events; complete calendar coverage does not guarantee equal administrative maturity across years. The absence of one historical state-month was handled explicitly, but unmeasured underrecording within available files remains possible. The pandemic may also have altered case mix, service use and coding; a national administrative curve cannot isolate its causal effect, and clinically selected COVID-19 cohorts are not directly comparable [22].

Finally, the analysis ends at discharge and cannot describe recovery, disability or later deaths.

Studies of post-near-miss follow-up emphasize outcomes beyond survival [23]. Administrative monitoring is therefore most useful for identifying patterns that warrant focused investigation, supporting transparent surveillance and directing questions to clinical and population-based studies.

Conclusion

SIH-SUS data from 2002–2025 identify regional and age variation in administrative markers of maternal near miss, with hypertensive diagnoses contributing prominently to the Waterstone-derived list. Age standardization preserved the higher pooled frequencies in the North, while requiring prenatal registration substantially changed the population and its regional comparisons. The historical Northeastern complement revealed variation in diagnostic composition and municipal associations across states. Socioeconomic associations depended on region, definition, temporal transformation and geographic scale. Clearly specified admission populations, diagnostic lists and denominators make these data useful for surveillance and hypothesis generation; explanations of clinical severity and contextual mechanisms require complementary evidence.

Data and analytical materials

The [public dashboard](#) provides aggregate tables and the manuscript, including the Northeastern supplementary tables listed below. The analytical materials in the [working repository](#) contain the codebook, source inventory, coverage matrix, age-standardization weights, all coefficient estimates and paired observations, and supplementary tables. The code operates on public DATASUS and IBGE data. Counts refer to admission authorizations; no person-level data are included in the dissemination dashboard.

Supplementary tables and figures

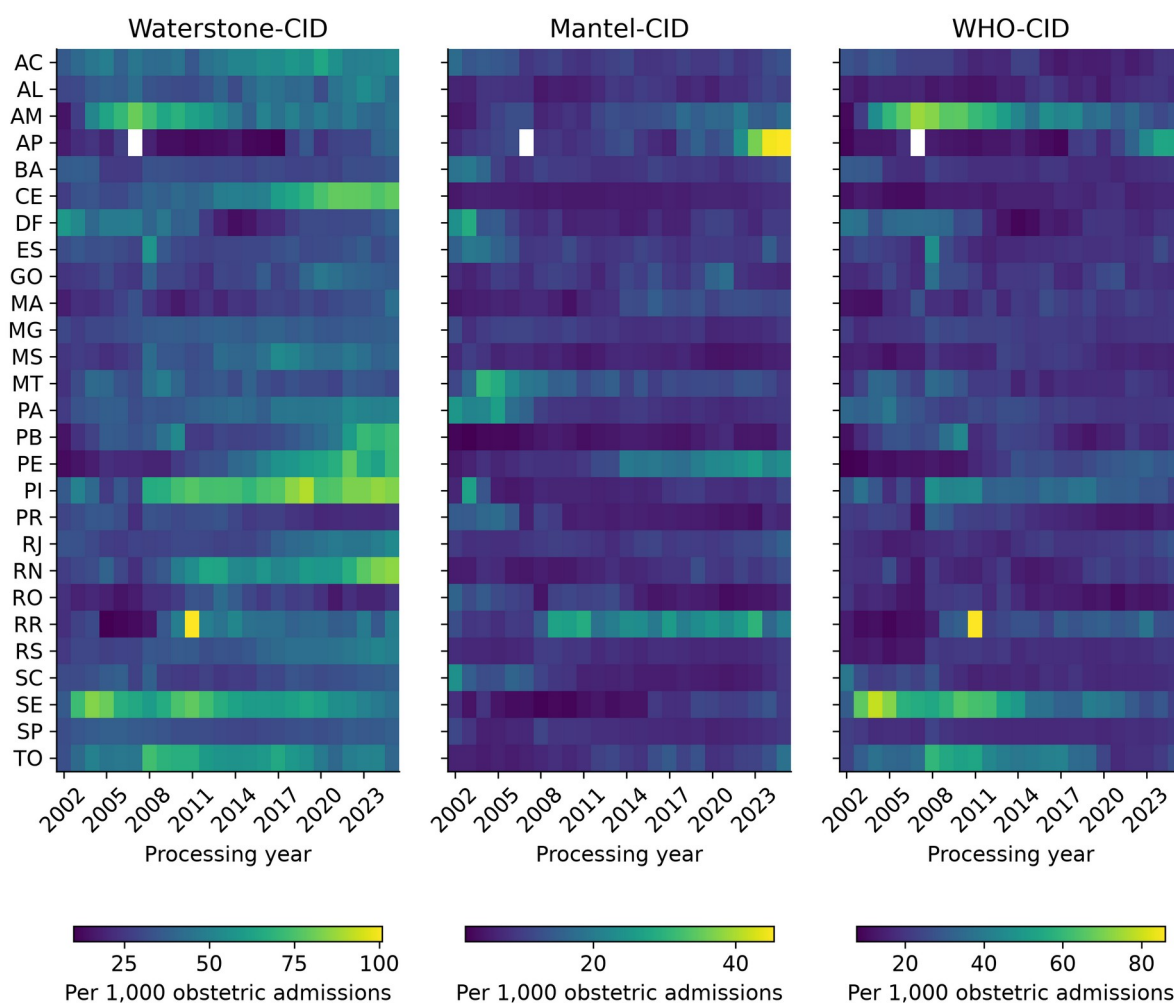
Supplementary Table S1. Rosendo-list identifications among Northeastern residents aged 15–49, 2008–2012. Hospitalizations in any Brazilian state; crude frequencies per 1,000 eligible obstetric admissions. Hypertension and hemorrhage are subsets of the overall marker set. NE denotes the pooled Northeast.

State	Admissions	Identifications	Overall / 1,000	Hypertensive / 1,000	Hemorrhagic / 1,000
AL	265,805	8,886	33.43	20.97	7.56
BA	1,023,955	32,248	31.49	14.66	7.68
CE	598,075	23,669	39.58	30.75	4.15
MA	504,485	11,740	23.27	8.86	9.99
PB	262,515	9,543	36.35	17.83	14.64
PE	637,464	16,008	25.11	14.15	2.67
PI	244,638	16,822	68.76	31.10	32.26
RN	221,669	10,329	46.60	34.16	6.93
SE	154,894	11,274	72.79	10.87	58.64
NE	3,913,500	140,519	35.91	18.91	10.59

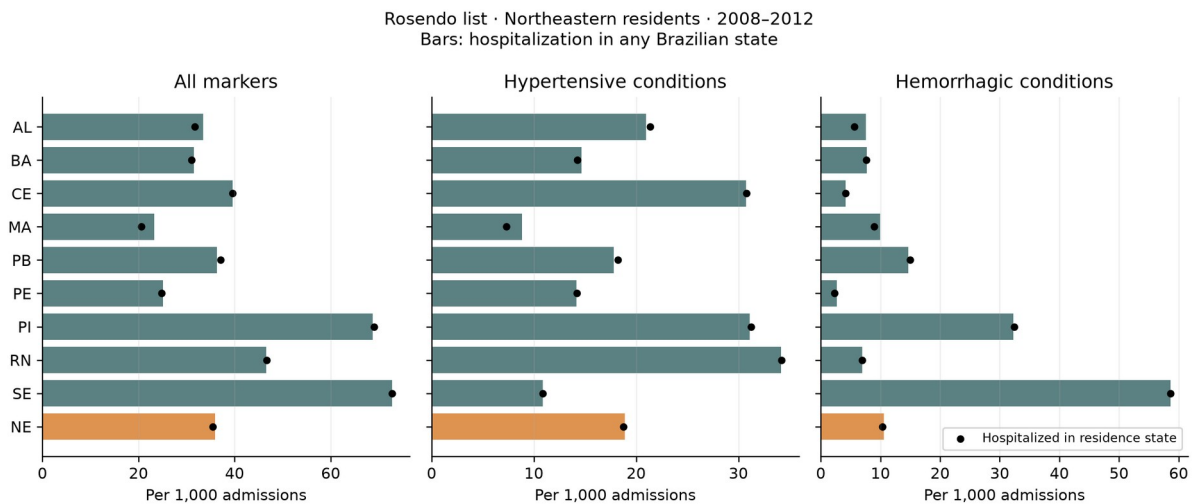
The complete machine-readable supplementary tables are available from the public dashboard:

- [Table S1: state totals and hospitalization-location sensitivity.](#)
- [Table S2: municipal counts, denominators, frequencies and Census covariates.](#)
- [Table S3: all 360 bivariate comparisons, conventional and block intervals, and separate Pearson/Spearman multiplicity adjustments.](#)
- [Table S4: correlations after subtracting state-specific means, including selection sensitivities.](#)
- [Table S5: exploratory principal-diagnosis composition by state and marker group.](#)

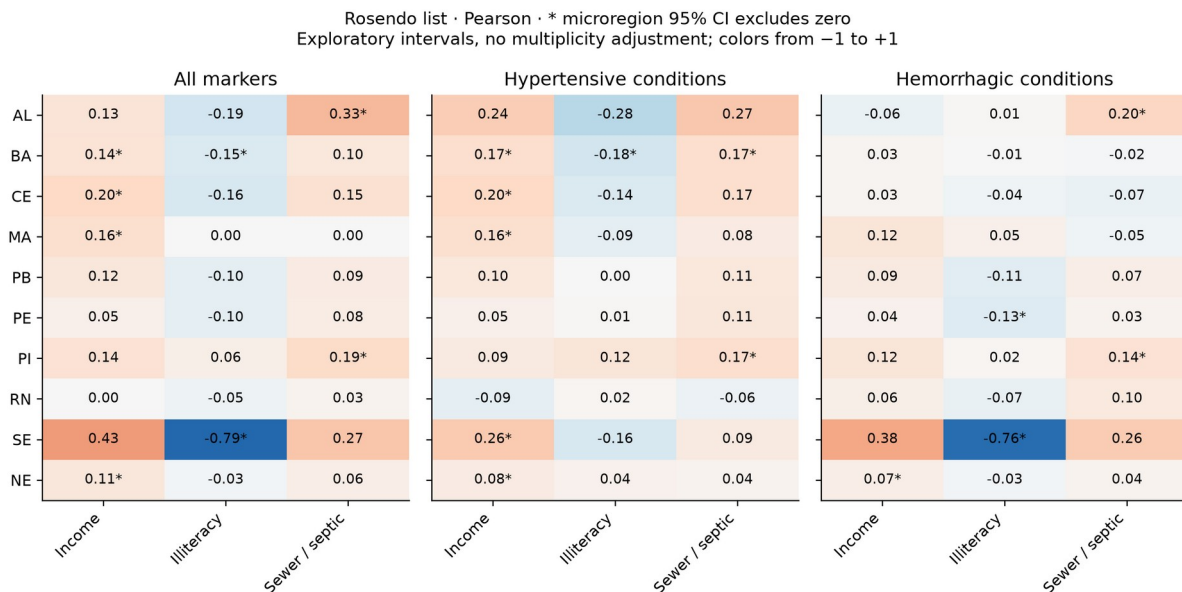
In these files, `care=any_uf` includes hospitalizations anywhere in Brazil and `care=same_uf` restricts hospitalization to the state of residence. `minimum_admissions=1` includes every municipality with an eligible admission; 1000 defines the volume sensitivity. *W* denotes the Rosendo marker set in this complement. Frequencies ending in `_per_1000` are crude. In Table S3, `pearson_q_bh` and `spearman_q_bh` adjust separate conventional test families; `block_ci_low` and `block_ci_high` delimit the unadjusted microregion-bootstrap interval.



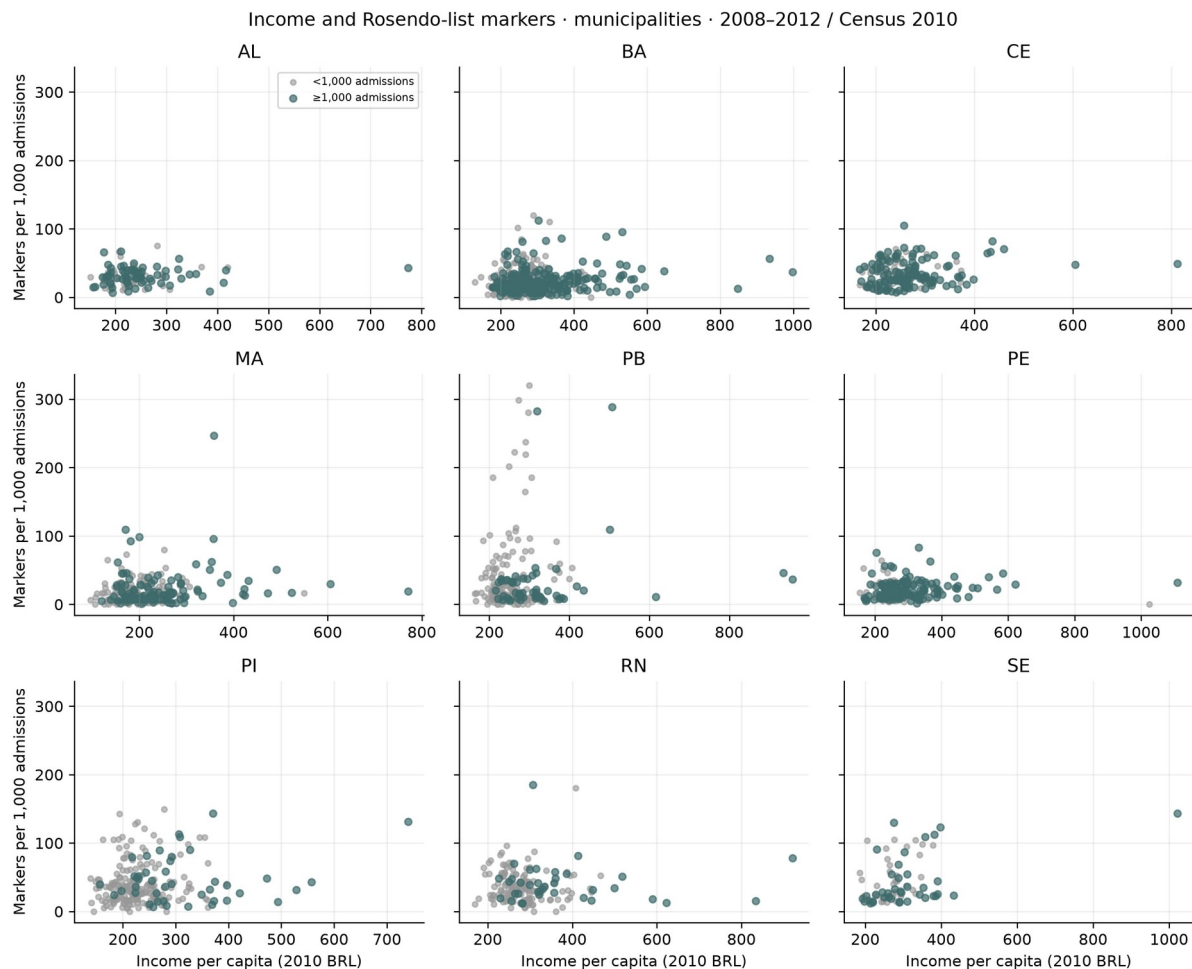
Supplementary Figure S1. State-level crude frequencies by processing year. Each panel has its own color scale. White cells indicate incomplete annual coverage, not zero frequency. AP/2007 is incomplete. These descriptive patterns do not adjust for referral flows or differences in recording.



Supplementary Figure S2. Crude Rosendo-list frequencies by state of residence and marker group, 2008–2012. Bars include hospitalizations anywhere in Brazil; black points restrict them to the state of residence. NE denotes the pooled Northeast. All denominators are eligible obstetric admissions among residents aged 15–49. Hypertensive and hemorrhagic identifications are subsets of the overall outcome.



Supplementary Figure S3. Municipal Pearson correlations by state, Census indicator and Rosendo-list outcome. All municipalities and hospitalizations anywhere in Brazil are included. Asterisks indicate microregion-bootstrap 95% intervals excluding zero, without multiplicity adjustment; they do not denote BH-adjusted findings. Colors range from –1 to +1. Full uncertainty estimates and both sets of q-values accompany Table S3.



Supplementary Figure S4. Municipal income per capita and overall Rosendo-list frequencies, Northeast, 2008–2012 / Census 2010. Each point is a municipality; green points have at least 1,000 admissions and gray points have fewer. Hospitalizations anywhere in Brazil are included. Frequencies are crude, and the panels do not represent the original study's spatial clusters. No fitted regression line or causal effect is implied.

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